

Exact Circulant-Block KRR on Regular-Grid Climate Fields: A CPU-Only Study

Anonymous Author(s) anon@example.org

Abstract. We study exact kernel ridge regression (KRR) on $H \times W$ regular-grid climate fields: the periodized (torus) kernel Gram matrix is a block-circulant-with-circulant-blocks (BCCB) matrix, so *circulant-block embedding and FFT spectral shrinkage* give closed-form training/prediction at $O(N \log N)$ time / $O(N)$ memory. We validate on **real data**: Kaplan SST v2 anomalies, 2005 monthly fields (1856-01..2023-01) on the 36×72 grid, future-blind train 1856-01..2015-12 / test 2016-01..2022-12. The spectral solve matches a dense floored-torus reference to $\sim 10^{-12}$ relative — exact for the *embedded-torus* kernel only; free-boundary gap 0.95 (matern32) / 0.12 (rbf). Masked reconstruction: matern32 RMSE 0.082–0.088 ($\sim 14\%$ of field std 0.60). Contribution: a CPU-only circulant-block implementation (78.7 s, 1.57 GB peak), a leakage-verified protocol, and a resource-accounted real-data benchmark; the pre-registered transfer-forecast claim is **REFUTED** (0/4); coverage violations disclosed per mask.

1 Introduction

Kernel ridge regression is a workhorse for spatial prediction in the AI-for-climate toolbox [5], but the $O(N^2)$ memory and $O(N^3)$ time of the dense solve bound its use on gridded climate fields. On *regular grids*, circulant embedding [7, 28] and FFT spectral shrinkage keep the kernel solve exact for the *embedded (torus) kernel* [11] (Section 2).

We contribute a CPU-only implementation, a leakage-verified protocol, and a resource-accounted real-data benchmark (Kaplan SST v2 [13]); we claim no FFT-kernel, spectral-regression, or ENSO-skill novelty; every number here is measured on real Kaplan SST v2 — never simulated.

2 Related work

Structured and randomized kernel methods. On regular grids the Gram of a periodized kernel is BCCB, and circulant embedding underpins *exact* simulation of stationary Gaussian fields [7, 28]; we make the *solve* exact via FFT spectral shrinkage [11]. Approximate alternatives trade exactness for scale: random Fourier features (RFF) [21], Nyström [26, 8], Fastfood [17], KISS-GP [27], GPyTorch [9], spectral-GP [16], SPDE/INLA [18]. The closest sibling is an *approximate* sketch-circulant KRR on non-grid data [29]; exactness is verified against dense references [7] (our P0 gate, Section 5.1: $\sim 10^{-12}$ relative).

Uncertainty quantification for dependent data. Split-conformal intervals rest on exchangeability; for serially dependent residuals we calibrate calendar-month blocks from the 60 terminal in-train years (non-exchangeable conformal theory [1], dependent-data variant [4], adaptive-conformal-inference (ACI) miscoverage updates [10]).

Significance for forecast comparisons. Predictive-accuracy comparisons on serially correlated errors use the Diebold–Mariano statistic [6], block resampling [15], and blocked cross-validation [3, 22, 24]; with only 7 yearly test blocks we report a year-block bootstrap MSE-difference CI instead of a Diebold–Mariano (DM)/HAC (heteroskedasticity- and autocorrelation-consistent) statistic.

ENSO forecasting baselines and ceilings. Simple linear inverse models [20] and the operational multi-model ensemble (NMME) skill record [2, 14] bound what a single-model transfer can claim, and the spring predictability barrier [25] makes skill start-month-dependent; recent deep/hybrid ENSO forecasters [12, 30] are GPU-scale. Our transfer arm (Section 5.5) is a leakage-verified *benchmark protocol* against persistence, climatology and AR(1) — not an ENSO-forecasting contribution.

3 Method: exact torus spectral KRR

Embedding and training. With circular row/column distances on the $H \times W$ grid, a stationary kernel k (Matern-3/2, RBF) yields a *BCCB* Gram matrix K determined by its first column c , diagonalized by the unitary 2D-DFT matrix U , $K = U \text{diag}(\hat{c}) U^*$ [11, 7]. Ridge regression

$\alpha = (K + \lambda I)^{-1}y$ reduces to $\alpha = \mathcal{F}_2^{-1}(\mathcal{F}_2(y)/(\hat{c} + \lambda))$ with $\hat{c} = \mathcal{F}_2(c)$ — two FFTs and one pointwise division, $O(N \log N) / O(N)$; spectral functionals use the same algebra (Section 5.4). **Spectral floor.** Non-torus kernels (RBF) give small negative BCCB eigenvalues; floored at 0 (PSD projection). **Torus exactness scope.** The BCCB Gram wraps the kernel across the grid boundary, so this spectral solve is exact for the *embedded-torus* kernel, *not* the free-boundary kernel; measured modeling gap per kernel on a real test-window field: 0.95 (matern32) / 0.12 (rbf) relative. **Masked training.** For genuinely masked training we solve the exact free-boundary system with preconditioned conjugate gradients (PCG; exact block-Toeplitz-with-Toeplitz-blocks (BTTB) matvec via the $(2N - 1)$ -mirror circulant, preconditioned by the floored-torus spectral inverse; tol 10^{-8} ; Section 5.3).

4 Data and evaluation protocol

Real data. NOAA PSL Kaplan SST v2 monthly anomalies [13]: 2005 fields, 1856-01..2023-01, 36×72 (5°) grid; netCDF-4 read with h5py; compute CPU-only (NumPy/SciPy/scikit-learn). **Missing data.** 53.4% of cells carry the sentinel -9.96921×10^{36} and are treated as missing (100% polar caps rows 0–5, 30–35; 73% high-N; 46% high-S; 14.6% tropics, land columns): imputed by the valid-cell monthly mean for training, excluded from *every* evaluation target. **Valid-only evaluation.** Every metric (RMSE, coverage, skill, correlation) uses *valid cells only*: denominators exclude imputed/missing cells — coverage = (valid in-threshold)/(valid in subset), never diluted by the 53.4% missing mass; conformity likewise. **Splits.** Fields are 0-indexed from 1856-01 (index 0). Future-blind: train [0, 1920) = 1856-01..2015-12; test [1920, 2004) = 2016-01..2022-12 (84 months); month 2004 (2023-01) excluded, so 1920+84+1=2005 fields in total. **Protocol (leakage-verified).** (i) masked-cell holdout on the test window: random 10/30/50% and halo ($w=1/2/4$) masks (a halo mask is the radius- w neighborhood of the persistent missing regions: polar caps and high-latitude strips); (ii) calendar-month-block split-conformal intervals (construction in Section 5.2), coverage diagnosed *per mask* on global, seam (2016–2017: the 24 test months at the train/test junction), and decay (2016: its first 12 months, where imputation-boundary effects decay) subsets, target 0.90 [1, 4, 10]; (iii) 5-fold *temporal-block* CV for the functional arm [3]; (iv) a future-blind forecast arm with a pre-registered win rule tested by year-block bootstrap [15] (7 blocks 2016–2022, 2000 resamples, seed 7); DM/HAC omitted at $n=7$ [6]; (v) pooled exact/Nyström/RFF baselines [21, 26, 19].

5 Experiments and results

5.1 P0: torus exactness on a real field

On a real test-window field (2016-01, index 1920, $\lambda=10^{-3}$) the spectral solve matches the dense floored-torus reference to 2.62×10^{-12} (matern32) / 1.12×10^{-12} (rbf) relative inf-norm — **PASS** at the 10^{-6} gate; torus-vs-free gap 0.95 / 0.12.

5.2 Reconstruction under leakage-aware masks (T1a)

On the 84 test months ($\lambda=10^{-4}$; Table 1): matern32 RMSE 0.0816–0.0880 ($\sim 14\%$ of field anomaly std 0.6001); rbf 0.2356–0.3155. Random-vs-halo gap +0.0040 (random 30% 0.0856 vs. halo $w=2$ 0.0816). Coverage (target 0.90): matern32 global 0.877–0.898, rbf 0.816–0.893; violations disclosed individually: matern32 halo $w=2$ decay 0.747, halo $w=4$ seam 0.836 / decay 0.739; rbf halo $w=2$ global 0.816 (seam 0.837), halo $w=4$ global 0.824. **Conformal construction.** Per kernel, conformity scores are $|residual|$ on *valid* cells of the 60 terminal in-train fields of the same calendar month (fields[1860:1920]); their 0.90 quantile is the per-month threshold; λ uses fields[1910:1920] only, test window untouched.

5.3 Genuinely masked training (T1b)

Exact free-boundary PCG (Section 3) at 10/30/50% observed converges to 10^{-8} in *all* runs: mean iterations 245–281 (span 233–291), final relative residuals 4.69×10^{-9} – 9.81×10^{-9} , masked RMSE 0.529–0.541; full-field T1a stays more accurate.

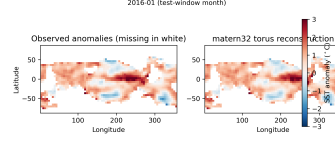


Figure 1: 2016-01 test-window field: observed anomalies (missing cells in white) vs. matern32 torus reconstruction ($\lambda=10^{-4}$, Section 5.2).

5.4 Nino3.4 functional index (T2)

The kernel-consistent functional (integral over the Nino3.4 box [23] via $K(\cdot, \mathbf{1}_{\text{box}})$) has RMSE 0.0856 (corr 0.997) under 5-fold temporal-block CV over all 2005 months, vs. 0.2696 for the direct ridge head on raw 2592-dim features ($3.2\times$) and 0.7955 for the zero model. **Scope note:** a leakage-safe *method* comparison (2016–2022 participate as CV folds), *not* a future-blind forecast claim (Figure 2).

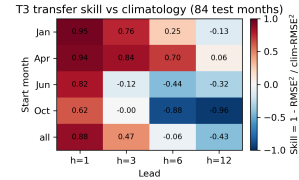


Figure 2: Left: Nino3.4 index (observed) vs. spectral box-functional prediction: the plotted curve is the 84-month test-window illustration of the train-fit functional ($\lambda=10^{-2}$); the quoted RMSE 0.0856 / corr 0.997 are the 5-fold temporal-block CV record over all 2005 months (Section 5.4). Right: T3 transfer skill vs. climatology by start month and lead ($n=7$ per origin).

5.5 Transfer forecast (T3): REFUTED under the pre-registered win rule

Pre-registered win rule: *supported* at h iff transfer beats persistence, climatology and AR(1) in RMSE and skill with a bootstrap-significant margin; *refuted* if $>$ half fail. **Aggregate verdict: REFUTED — 0/4 horizons supported.** In Table 2 (skill = $\text{MSSS} = 1 - \text{MSE}/\text{MSE}_{\text{clim}}$; negative = worse than train-only climatology), at $h=1$ the transfer (0.268) loses to persistence (0.251) and AR(1) (0.241); at $h=3$ it loses to AR(1) (0.508); at $h=6$ it beats persistence (0.796 vs. 0.907) but the MSE-difference CI includes zero ($p=0.234$, se 0.197); at $h=12$ skill is -0.427 (floor). **Minimum detectable margin.** With 7 year blocks and 2000 draws, the bootstrap CI half-width (observed se 0.0056–0.333 across comparisons, vs-persistence 0.007–0.333) sets the minimum detectable MSE margin; *supported* requires the margin to exceed that half-width at the 95% level — the closest margin, $h=1$, is -0.0137 (transfer loses to AR(1)), not significant. Truncation sweep (2012/2013/2014–12 cutoffs): $h=1$ skill 0.8795–0.8797.

Skill is origin-dependent (Table 3, Figure 2): January best at $h=1$ (0.954); April strongest at $h \geq 3$ (0.84/0.70/0.06); October worst over $h=1,6,12$ (0.62 / -0.00 / -0.88 / -0.96 across $h=1,3,6,12$; June worse at $h=3$); by $h=12$ every origin is at or below the climatology floor — only April is marginally positive (0.059).

5.6 Solve-cost control on a fixed pooled-cell task (B)

On the *same* 2000 valid test cells, per-field spectral reconstruction has pooled RMSE 0.0999; pooled exact/Nyström/RFF KRR on (lat,lon) coordinates reach 0.820–0.824 (Table 4). Coordinate-only baselines sit at field-std level (zero 0.801 / global mean 0.605 / per-cell climatology 0.813) — a *solve-cost control at a fixed task*, not an end-to-end comparison.

5.7 Resolution sensitivity (R) and scaling (S)

2×2 coarsening to 10° degrades reconstruction RMSE 0.0853 to 0.1667 (84 fields) and functional RMSE 0.0856 to 0.3825 (corr 0.997 to 0.950) — an honest negative. Fits are sub-ms through

$N=10,368$ (0.0002–0.0005 s at $N \in \{1024, 2592, 10368\}$), $O(N \log N)$ flops / $O(N)$ bytes; a 0.25° field ($N=1,036,800$ cells; one float64 field is 8.3 MB) fits in ~ 0.04 GB — the five working arrays of one spectral solve: input field y , kernel first column c , rfft2 spectrum, dual coefficients α , and the pointwise $(\hat{c} + \lambda)$ buffer. **Resources (CPU-only):** 78.7 s wall-clock, 1.57 GB peak RSS (B dominates — Nyström-5000 at 29.8 s).

Mask	RMSE m32	cov m32	RMSE rbf	cov rbf
random 10%	0.0843	0.898	0.2525	0.873
random 30%	0.0856	0.892	0.2411	0.893
random 50%	0.0872	0.886	0.2356	0.892
halo $w=1$	0.0827	0.896	0.2789	0.856
halo $w=2$	0.0816	0.897	0.2839	0.816
halo $w=4$	0.0880	0.877	0.3155	0.824

Table 1: T1a (84 test months): masked-cell RMSE and coverage (target 0.90) per mask; violations in Section 5.2, none averaged.

h	transfer	persistence	climatology	AR(1)	skill
1	0.268	0.251	0.773	0.241	0.880
3	0.565	0.562	0.773	0.508	0.466
6	0.796	0.907	0.773	0.766	−0.061
12	0.923	1.154	0.773	0.879	−0.427

Table 2: T3 transfer RMSE by lead (84 test months; skill = MSSS vs. train-only climatology: 0.880/0.466/−0.061/−0.427; negative = worse than climatology). **Verdict: REFUTED** — 0/4 horizons.

Origin	$h=1$	$h=3$	$h=6$	$h=12$
Jan	0.954	0.760	0.248	−0.126
Apr	0.937	0.844	0.697	0.059
Jun	0.816	−0.120	−0.437	−0.318
Oct	0.620	−0.001	−0.884	−0.955
all	0.880	0.466	−0.061	−0.427

Table 3: T3 start-month $\times$ lead skill vs. climatology ($n=7$ per origin); at $h=12$ every origin is at or below the climatology floor except April (+0.059).

Method	RMSE	flop ratio	acc/flop (orders)
spectral (2592)	0.0999	1	—
ridge $n=2000$	0.824	5.3×10^4	5.6
ridge $n=4828$	0.820	7.5×10^5	6.8
Nyström $k=5000$	0.820	1.6×10^6	7.1
RFF 5000	0.822	4.8×10^2	3.6

Table 4: B: pooled-cell RMSE (2000 cells) and solve cost vs. spectral; acc/flop (orders) = $-\log_{10}(\text{accuracy-per-flop vs. spectral})$, so higher = worse; baselines 0.801/0.605/0.813 — solve-cost control on a near signal-free task.

6 Limitations and discussion

(i) **Exactness is torus-embedded only:** free-boundary gap 0.95 (matern32) / 0.12 (rbf); masked training falls back to PCG at $\sim 6\times$ the T1a error. (ii) **T3 transfer superiority is REFUTED** (0/4): 1-month skill reflects autocorrelation already captured by persistence/AR(1); negative floor at $h=12$; no transfer or ENSO-skill claim. (iii) **Coverage is not uniform:** the reported-violation range on seam/decay subsets is 0.739–0.837 (over the reported $w=2/w=4$ halo masks: matern32 $w=4$ decay 0.739 / seam 0.836; rbf $w=2$ seam 0.837); all violations reported, none averaged. (iv) **Resolution sensitivity** is an honest negative (R/S above). (v) **Kaplan smoothing caveat:** EOF-based reconstruction [13]; missing cells (53.4%) imputed by monthly means, excluded from all targets.

Conclusion

Exact circulant-block KRR is a practical CPU-only route on regular grids: torus-exact closed-form solves at $O(N \log N) / O(N)$ (78.7 s / 1.57 GB); masked reconstruction at $\sim 14\%$ of field std; the pre-registered transfer claim is refuted, coverage violations and degradation reported — the contribution stands on implementation, protocol, benchmark, not novelty.

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